

CENTRAL BOARD EXAMINATION - 2023-2024

CLASS 12 — MATHEMATICS

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains 10 questions divided into Section A and Section B.
2. Section A contains 5 Multiple Choice Questions carrying 1 mark each.
3. Section B contains short & long answer questions carrying 3 to 5 marks each.
4. All questions are compulsory. Read the diagrams carefully where provided.

SECTION A (Multiple Choice Questions)

Q1. What is the SI unit of electric current?

- (a) Volt (b) Ampere (c) Ohm (d) Joule

Q2. Which of the following is a balanced chemical equation?

- (a) $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$ (b) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$ (c) $\text{H}_2 + 2\text{O}_2 \rightarrow \text{H}_2\text{O}$

Q3. The focal length of a spherical mirror is 20 cm. Its radius of curvature is:

- (a) 10 cm (b) 20 cm (c) 40 cm (d) 5 cm

Q4. Which gas is evolved when dilute hydrochloric acid reacts with zinc granules?

- (a) Oxygen (b) Hydrogen (c) Carbon dioxide (d) Nitrogen

Q5. The speed of light in vacuum is approximately:

- (a) $3 \times 10^8 \text{ m/s}$ (b) $3 \times 10^5 \text{ m/s}$ (c) $1.5 \times 10^8 \text{ m/s}$ (d) $3 \times 10^6 \text{ m/s}$

SECTION B (Short & Long Answer Questions)

Q6. State Ohm's Law and write its mathematical formula.

- (a) Define resistance of a conductor.
(b) What factors affect the resistance of a wire?

Q7. Differentiate between autotrophic and heterotrophic nutrition with examples.

Q8. State the law of conservation of energy with an example.

Q9. Describe an experiment to demonstrate chemical reactions.

Q10. Explain the process of photosynthesis with a balanced chemical equation.